MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MG5DSCECT302 - Embedded Systems with AVR Microcontroller

PRACTICAL EXAMINATION

Duration: 2 hours Maximum Marks: 35

Instructions:

- Demonstrate the given task (Any one of the tasks listed below)
- Viva session: (Minimum 5 questions from the practical mode)
- Lab Report

Evaluation Criteria:

- Viva: 10 marks
- Lab Report: 5 marks
- Demonstration: 20 marks

Tasks: (Select any two, one each from the interfacing and non-interfacing experiment category)

A. Non Interfacing

1. 8 bit addition, subtraction
2. Multiplication with and without using MUL instruction
3. Division
4. Ascending order
5. Descending order
6. Find out largest number
7. Find out smallest number
8. Swapping
9. Count number of ones in a byte
10. Find out odd and even numbers in an array

B. Interfacing programs

11. Stepper motor interface
12. DC motor interface
13. Keyboard interface
14. LCD interface
15. PWM
16. LED interface and Delay programming
17. Buzzer interface
18. Relay interface
19. Timer programming
20. Optoisolator interface
21. DS1307 RTC interfacing
22. RGB LED Interfacing